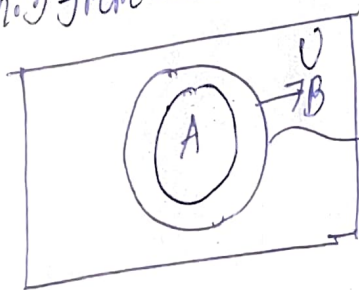


By Defn. 1, $(x \in A \vee x \in B) \wedge (x \in A \vee x \in \bar{A})$. By 1st Distributive Law, of Propositions, $(x \in A \vee (x \in B \wedge x \in \bar{A}))$. By Defn. 4, $(x \in A \vee x \in (B \wedge \bar{A}))$. By Defn. 1, $(x \in A \vee (x \in B \wedge x \in \bar{A}))$. By Defn. 4, $(x \in A \vee x \in (B \wedge \bar{A}))$. By Defn. 1, $x \in (A \vee (B \wedge \bar{A}))$. By Defn. 3 from Section 2.1, $A \cup B \subseteq A \cup (B \wedge \bar{A})$.
 $\therefore A \cup (B \wedge \bar{A}) = A \cup B \quad \square$

17) Show that if A and B are sets in a universe U then $A \subseteq B$ if and only if $A \cup B = U$.

Solution: Given, $A \subseteq U$ and $B \subseteq U$.
 Let, $A \subseteq B$. Let, x be an arbitrary element of A , i.e. $x \in A$.
 By Defn. 3 from Section 2.1, $x \in B$. By Defn. 3 from Section 2.1, $x \in U$.



$\rightarrow A \subseteq B$, but $A \cup B \neq U$

17) Show that if A and B are sets in a universe U then $A \subseteq B$ if and only if $\bar{A} \cup B = U$.

Solution: Let, $\bar{A} \cup B = U$. $\therefore \bar{A} \cup B \subseteq U$ and $U \subseteq \bar{A} \cup B$. It's given that $A \subseteq U$ and $B \subseteq U$. By Prob-19 from Exercise 2.1, since $A \subseteq U$ and $U \subseteq \bar{A} \cup B$, $\therefore A \subseteq \bar{A} \cup B$.

Let, x be an arbitrary element of A , i.e. $x \in A$. By Defn. 3 from Sec-2.1, $x \in \bar{A} \cup B$. By Defn. 1, $x \in \bar{A} \vee x \in B$. By Defn. 5, $x \notin A \vee x \in B$. By Defn. of not belongs to, $\neg(x \in A) \vee x \in B$. By Disjunctive syllogism, $x \in B$.
 $\therefore$ From Defn. 3 of Section 2.1, $A \subseteq B$.

Let, $A \subseteq B$. $\therefore B \subseteq U$. If x is an arbitrary element of B , i.e. $x \in B$, from Defn. 3 of Section 2.1, $x \in U$. By Defn. of OR, $(x \in \bar{A} \vee x \in B)$ is True.
 By Defn. of conditional, $(x \in \bar{A} \vee x \in B) \rightarrow (x \in U)$ is True. From Defn. 1,